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Water Wave Simulation – GPU-Based Fluid Dynamics and Procedural Ocean System

Water Wave Simulation (GPU Fluid Simulation Layer)

System Overview

This module transforms water rendering into a fully dynamic physics-based simulation system executed entirely on the GPU. Instead of treating water as a static surface or animated texture, it is defined as a mathematical wave field that evolves continuously over time.

The core principle is:

Water is not rendered as geometry motion. It is simulated as a continuous mathematical wave system.

https://www.udemy.com/course/web-based-3d-terrain-industrial-printing-engine-mastery/

Sin and Cos Wave Displacement System

Mathematical Wave Model

Water surface motion is generated using trigonometric functions in the vertex shader.

$y(x, t) = A \cdot \sin(kx - \omega t)$

Where:

- A defines wave amplitude

- x defines spatial position
- t defines time progression

Wave Layer Composition

Primary Sin Wave Layer

Represents large-scale ocean swell and dominant wave motion.

Secondary Cos Wave Layer

Adds fine surface oscillation and micro variation.

Superposition System

Multiple wave functions are combined to produce complex natural motion behavior.

Engineering Outcome

- Water surface becomes fully procedural
- No geometry updates required
- All motion is computed in GPU vertex shader stage
- Scalable to large environments

Wind-Driven Wave System

Vector-Based Flow Control

Wind is treated as a directional force field that influences wave behavior.

w = (wx, wy)

Where:

- wx controls horizontal wave propagation
- wy influences vertical displacement behavior

Wind Parameters

Direction Control

Determines the directional flow of wave energy across the surface.

Modulates wave amplitude and intensity.

Choppiness Factor

Introduces irregular surface instability for realism.

Engineering Outcome

- Water motion becomes environment-dependent
- Wave propagation is dynamically controlled
- Physical simulation replaces keyframe animation

Multi-Frequency Wave Blending System

Fractal Wave Composition Model

Real ocean surfaces consist of overlapping wave systems at different scales.

$W(x, t) = \Sigma A_i \cdot \sin(k_i x - \omega_i t)$

Where:

- A_i represents amplitude per layer
- k_i defines wave frequency scale
- ω_i controls time-based evolution

Frequency Layers

Low Frequency Layer

Large-scale ocean swells and long wave systems.

Mid Frequency Layer

Medium-scale surface deformation and variation.

High Frequency Layer

Fine ripples and micro surface noise.

System Behavior

- Noise injection improves realism

Engineering Outcome

- Fractal-like wave structure
- Multi-layer shader-based simulation
- High realism without geometry complexity

Foam Edge Detection System

Depth-Based Interaction Model

Foam is generated where water interacts with terrain boundaries.

F(x) = max(0, dterrain − dwater)

Where:

- dterrain is terrain depth reference
- dwater is water surface depth

Foam Generation Logic

Depth Buffer Comparison

Water surface is compared against terrain elevation to identify collision zones.

Threshold Detection

Shallow intersections generate foam regions.

Noise Modulation

Random variation is applied to avoid artificial uniformity.

Engineering Outcome

- Real-time shoreline simulation
- Physically consistent wave breaking
- GPU-based edge detection system

System-Level Fluid Simulation Architecture

Before This Module

- Static water plane
- Texture-based animation
- Fake wave illusion

After This Module

- Procedural wave function system
- Wind-driven physical simulation
- Multi-frequency dynamic surface model
- Depth-aware interaction system

Engine Interpretation Model

At system level:

- Water is a mathematical field
- Waves are sinusoidal time functions
- Wind is a vector force input
- Foam is a collision-derived surface effect

This establishes a unified GPU fluid simulation system.

Final Engineering Tasks

Task 1 — Wave Function Shader System

Design a vertex shader that:

- Implements sine-based wave displacement
- Supports multi-layer frequency stacking
- Allows real-time amplitude control

Explain time-based deformation behavior.

Task 2 — Wind Influence System

Design a system that:

- Uses a 2D wind vector field

Explain vector influence on wave propagation.

Task 3 — Multi-Frequency Wave Engine

Design a system that:

- Combines at least three wave frequencies
- Applies phase offsets between layers
- Generates realistic ocean surface motion

Explain why single-frequency models fail.

Task 4 — Foam Generation System

Design a system that:

- Uses depth buffer comparison
- Generates shoreline foam dynamically
- Applies noise-based variation

Explain GPU depth testing contribution.

Engine Progression Context

After this module, the system includes:

- GPU terrain system
- Biome and vertex painting systems
- Non-destructive editing architecture
- Real-time fluid simulation layer
- Environmental interaction systems

Next Module Evolution

The next stage introduces:

- Volumetric cloud simulation
- Atmospheric density modeling
- Weather state machines

Final Engineering Statement

Water is not a rendered object.

It is a continuously evaluated mathematical system where wave functions, wind forces, and terrain interactions converge to produce a fully simulated real-time fluid surface.

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